西北工業大學
NORTHWESTERN POLYTECHNICAL UNIVERSITY

$\pi_0 \rightarrow \pi_{0.5} \rightarrow \pi_{0.6}$

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What's IndoorUAV?

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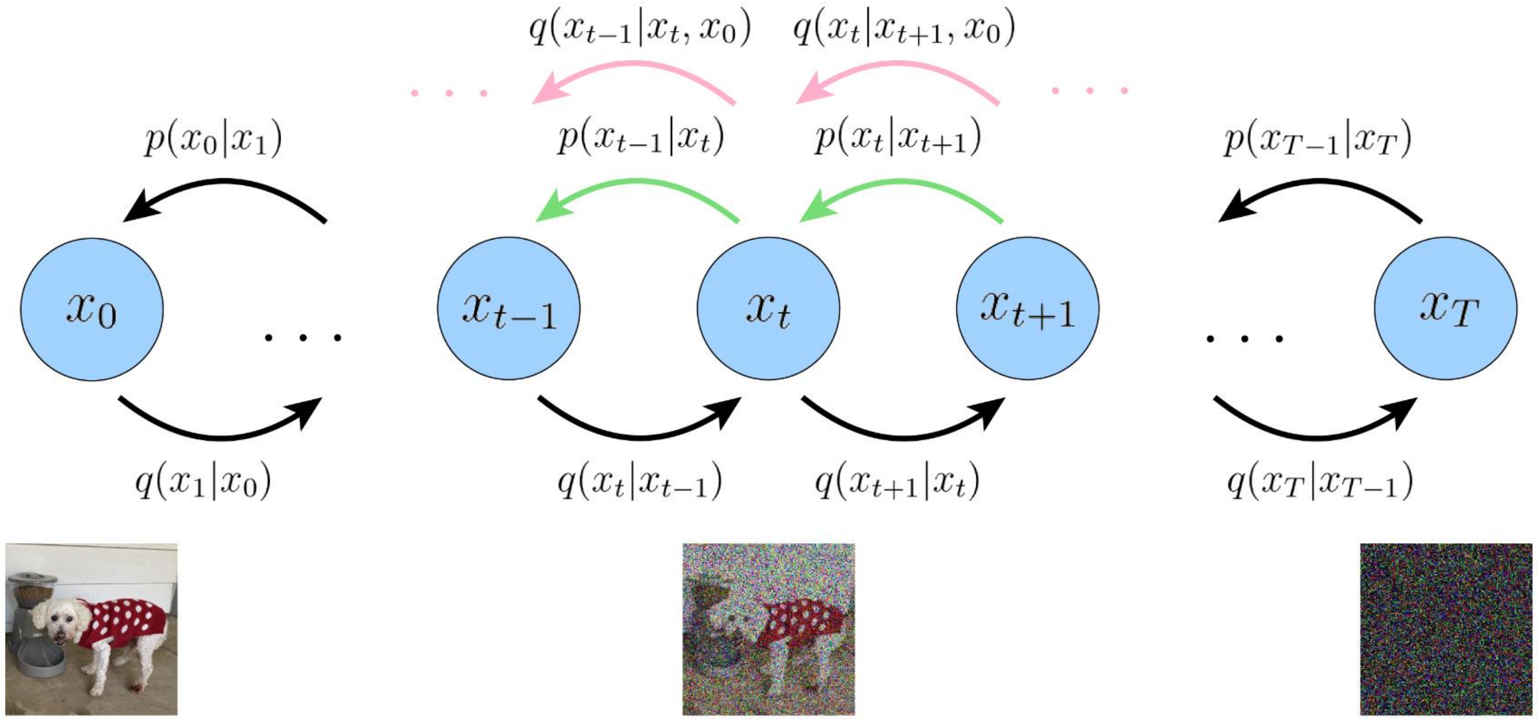
2026.02.05

Denoising Diffusion Probabilistic Models



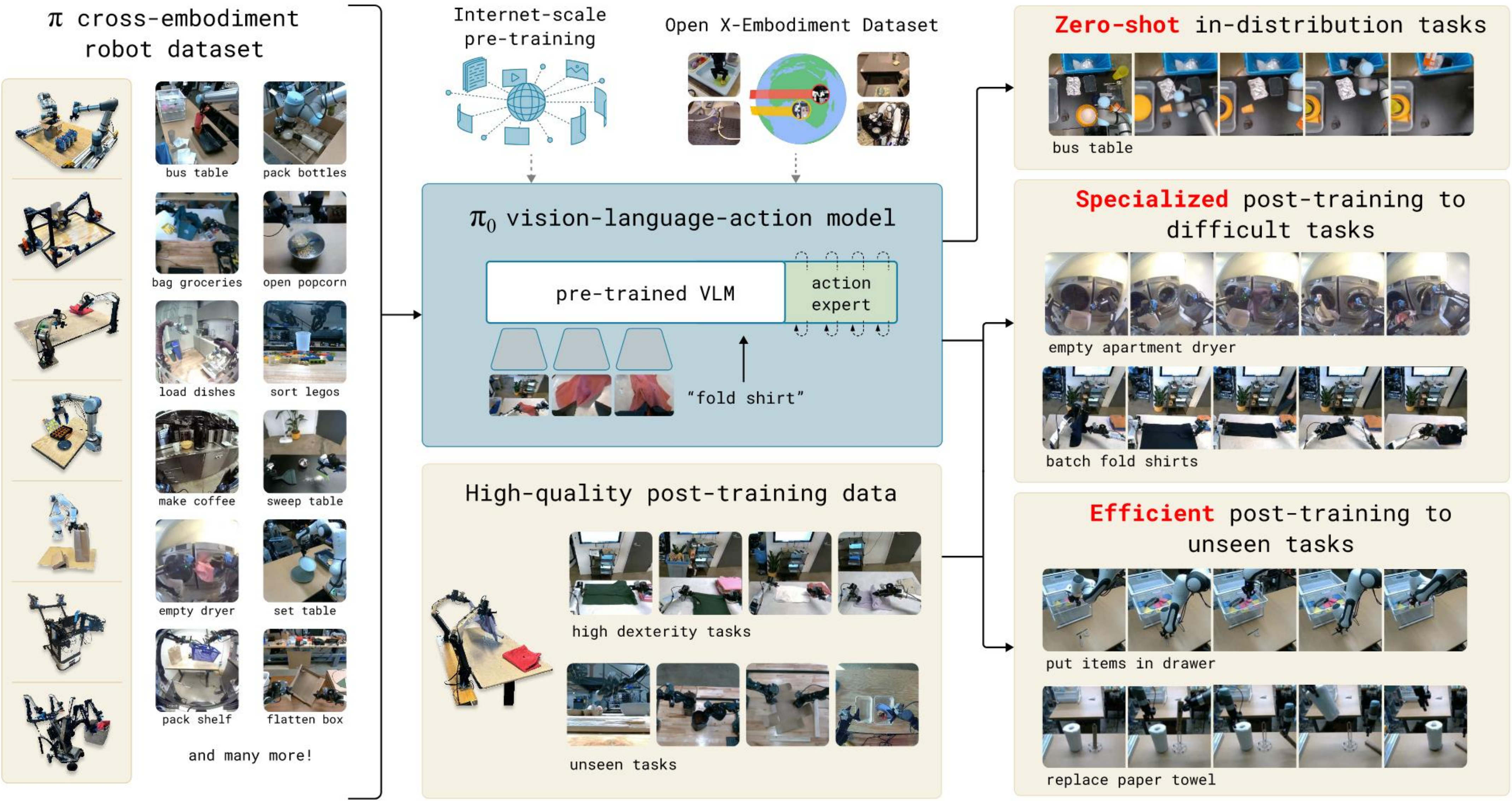
$$x_0, x_t$$

$$\downarrow$$

$$q(x_{t-1}|x_t, x_0)$$

$$q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, \boldsymbol{x}_0) = \frac{q(\boldsymbol{x}_t|\boldsymbol{x}_{t-1}, \boldsymbol{x}_0)q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_0)}{q(\boldsymbol{x}_t|\boldsymbol{x}_0)}$$

π_0: A Vision-Language-Action Flow Model for General Robot Control



only pre-training

high-quality post-training with larger datasets

efficient post-training with small to moderate amounts of data

Training

Training example

$$\mathbf{o}_t = [\mathbf{I}_t^1, \dots, \mathbf{I}_t^n, \ell_t, \mathbf{q}_t]$$

$$\mathbf{A}_t = [\mathbf{a}_t, \mathbf{a}_{t+1}, \dots, \mathbf{a}_{t+H-1}]$$

Random sampling

$\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \tau \in [0,1]$

Computing noisy actions $\mathbf{A}_t^\tau = \tau \mathbf{A}_t + (1-\tau)\epsilon$

Computing denoising vector

$$\mathbf{A}_t - \mathbf{A}_t^\tau = (1-\tau)(\mathbf{A}_t - \epsilon)$$

? $\mathbf{u}(\mathbf{A}_t^\tau | \mathbf{A}_t) = \mathbf{A}_t - \epsilon$

Network outputs

$$\mathbf{v}_\theta(\mathbf{A}_t^\tau, \mathbf{o}_t)$$

Matching

$$\boxed{L^\tau(\theta) = \mathbb{E}_{p(\mathbf{A}_t|\mathbf{o}_t), q(\mathbf{A}_t^\tau|\mathbf{A}_t)} ||\mathbf{v}_\theta(\mathbf{A}_t^\tau, \mathbf{o}_t) - \mathbf{u}(\mathbf{A}_t^\tau|\mathbf{A}_t)||^2}$$

Task used: H=50

Inference

$$\mathbf{A}_t^\tau = \tau \mathbf{A}_t + (1-\tau)\epsilon$$

$$\mathbf{A}_t^0 = \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \xrightarrow[\text{Denoising}]{\tau: 0 \to 1} \mathbf{A}_t^1 = \mathbf{A}_t$$

Euler integration

$$\mathbf{A}_t^{\tau+\delta} = \mathbf{A}_t^\tau + \delta \mathbf{v}_\theta(\mathbf{A}_t^\tau, \mathbf{o}_t)$$

Discussion on choice of δ:

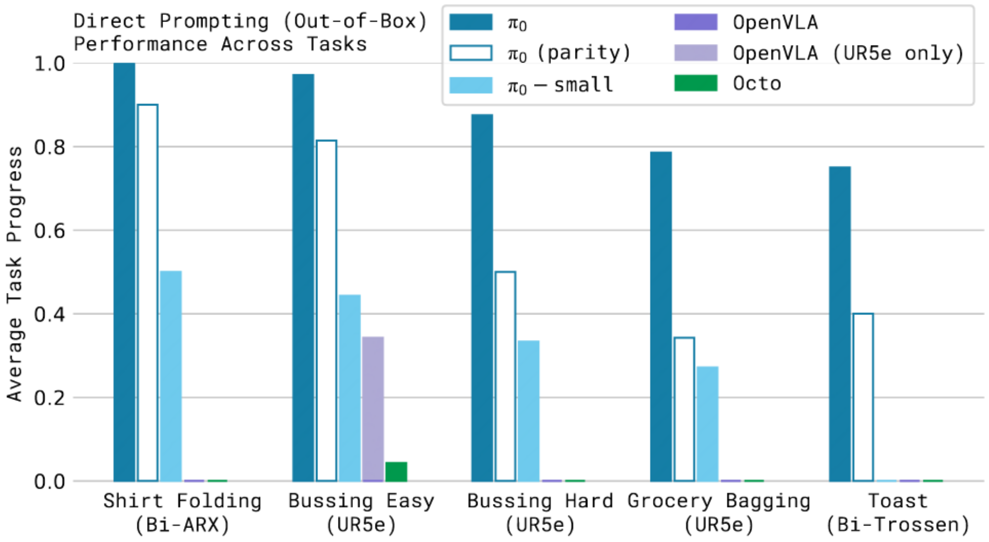
The Euler integration rule is essentially based on first-order Taylor expansion, with a local truncation error of $O(\delta^2)$

large $\delta \to$ overshooting and increased integration error

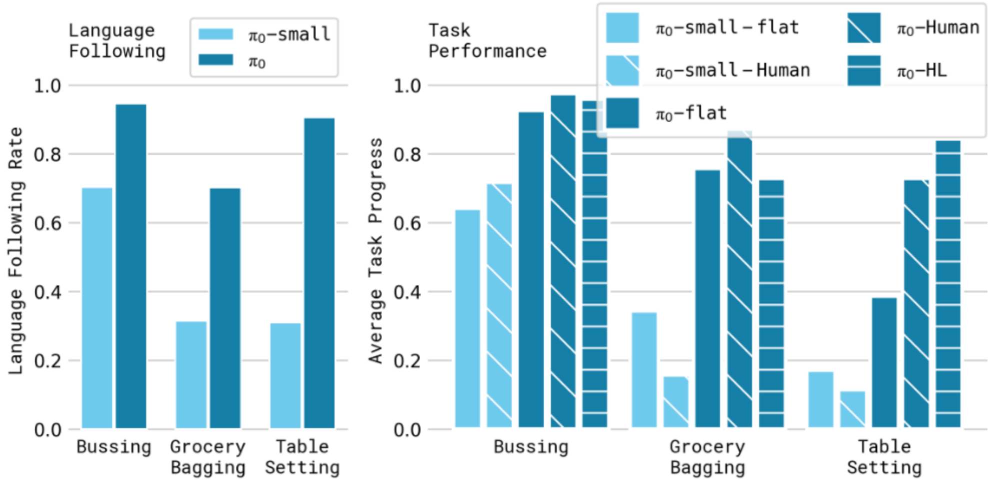
small $\delta \to$ high computational cost

Task used: δ=0.1

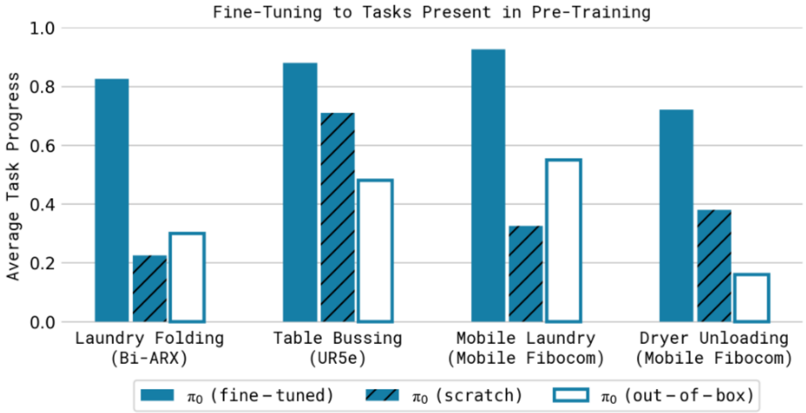
Evaluation

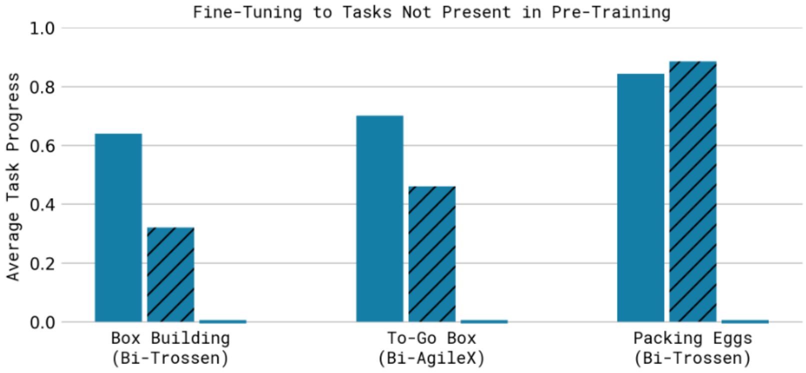


π_0: VLM + 700k steps π_0 (parity): VLM + 160k steps π_0 – small: without VLM

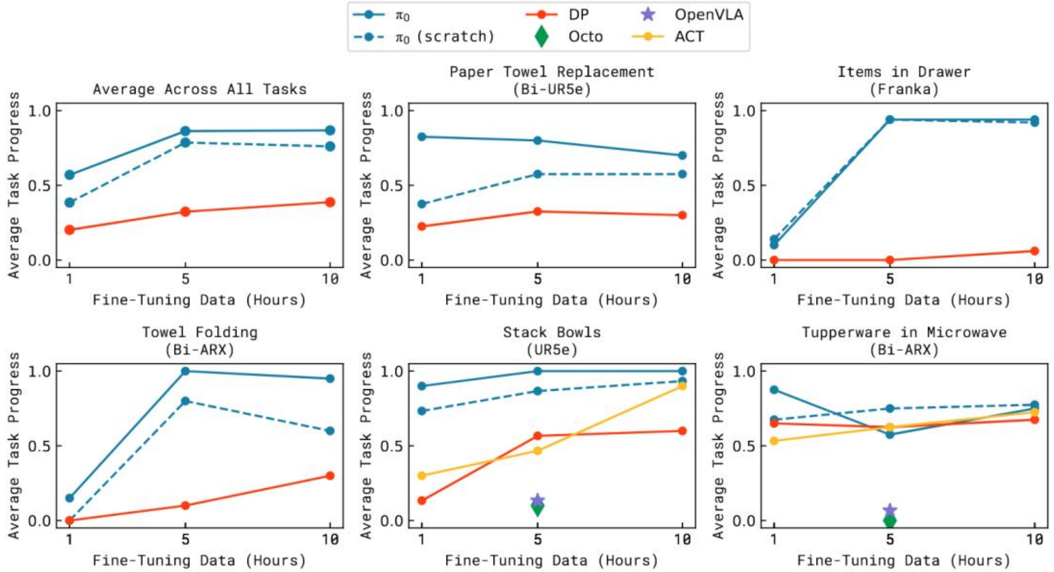


flat: task description Human: intermediate step commands from an expert human user
HL: high-level commands provided by a high-level VLM





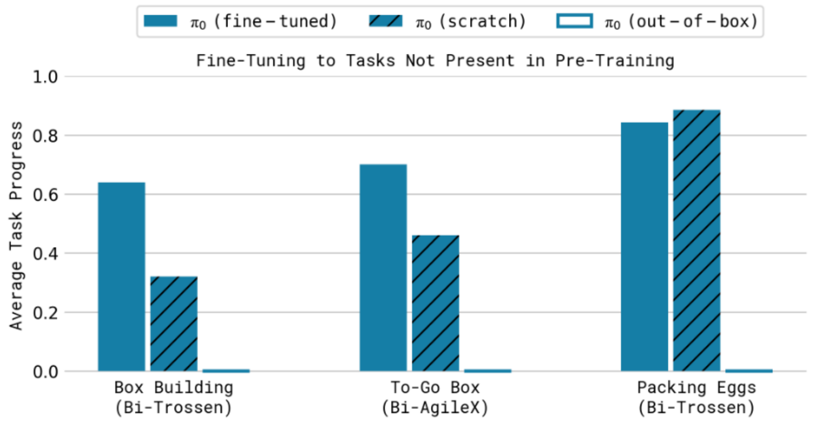
out-of-box: only pre-training
scratch: only fine-tuning
fine-tuned: pre-training + fine-tuning



π $\pi_{0.5}$: a Vision-Language-Action Model with Open-World Generalization

Why?

How?



The generalization performance of the only pre-trained π_0 is very poor.

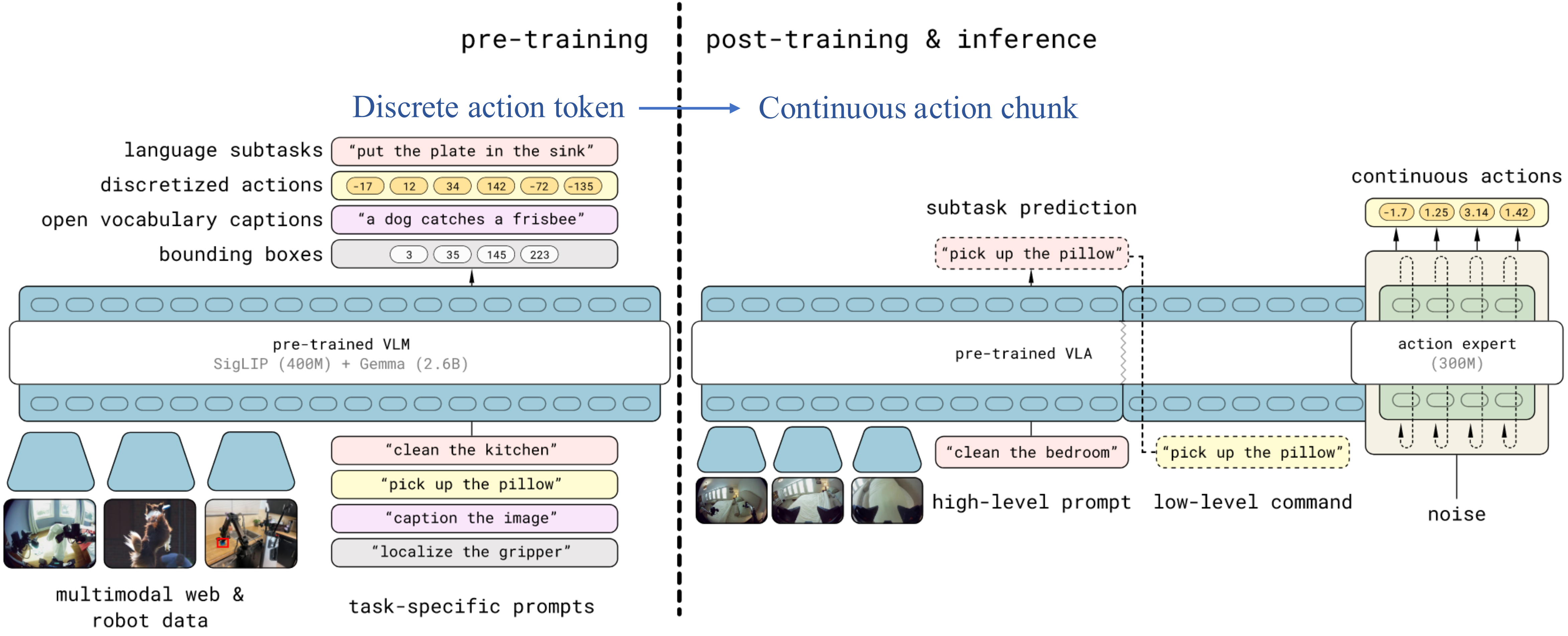
- Generalize readily if tasks are well presented in the data
- require adapting or modifying existing skills
- require understanding the semantics of the scene based on prior knowledge

How?

co-training
hybrid multi-modal examples

Transfer from a variety of information resources

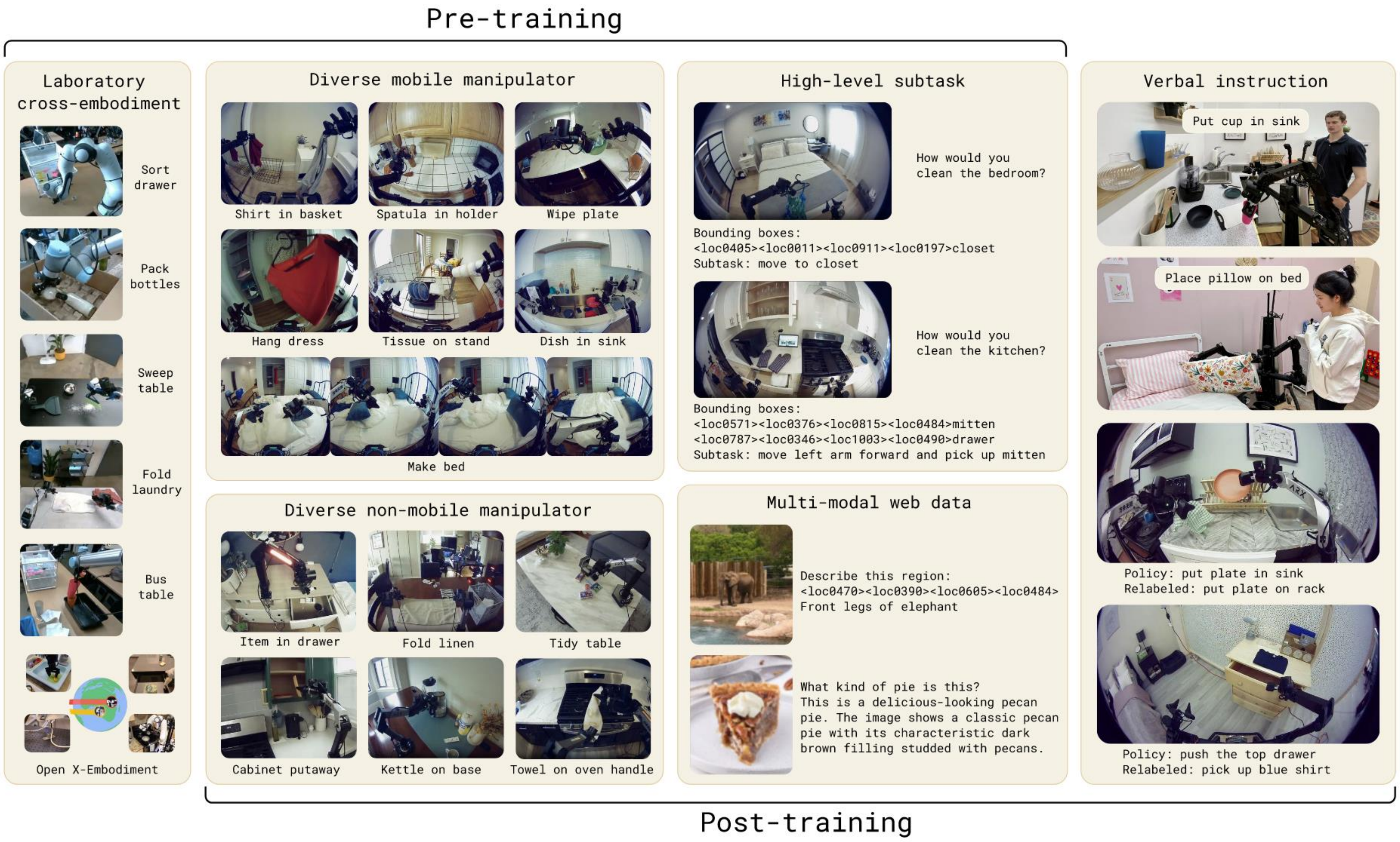
Model architecture



$$\pi_\theta(\mathbf{a}_{t:t+H}, \hat{\ell} | \mathbf{o}_t, \ell) = \underbrace{\pi_\theta(\mathbf{a}_{t:t+H} | \mathbf{o}_t, \hat{\ell})}_{\text{Low-level action inference}} \underbrace{\pi_\theta(\hat{\ell} | \mathbf{o}_t, \ell)}_{\text{High-level semantic inference}}$$

$$\mathbb{E}_{\mathcal{D},\tau,\omega}\Big[\underbrace{H(x_{1:M}, f_\theta^\ell(\mathbf{o}_t, \ell))}_{\text{Semantic inference}} + \alpha \underbrace{\left\| \omega - \mathbf{a}_{t:t+H} - f_\theta^a(\mathbf{a}_{t:t+H}^{\tau,\omega}, \mathbf{o}_t, \ell) \right\|^2}_{\text{Flow matching}} \Big]$$

Training


Pre-training
Laboratory cross-embodiment
Sort drawer
Pack bottles
Sweep table
Fold laundry
Bus table
Open X-Embodiment
Diverse mobile manipulator
Shirt in basket
Spatula in holder
Wipe plate
Hang dress
Tissue on stand
Dish in sink
Make bed
Diverse non-mobile manipulator
Item in drawer
Fold linen
Tidy table
Cabinet putaway
Kettle on base
Towel on oven handle
High-level subtask
How would you clean the bedroom?
Bounding boxes:
<loc0405><loc0011><loc0911><loc0197>closet
Subtask: move to closet
How would you clean the kitchen?
Bounding boxes:
<loc0571><loc0376><loc0815><loc0484>mitten
<loc0787><loc0346><loc1003><loc0490>drawer
Subtask: move left arm forward and pick up mitten
Multi-modal web data
Describe this region:
<loc0470><loc0390><loc0605><loc0484>
Front legs of elephant
What kind of pie is this?
This is a delicious-looking pecan pie. The image shows a classic pecan pie with its characteristic dark brown filling studded with pecans.
Verbal instruction
Put cup in sink
Place pillow on bed
Policy: put plate in sink
Relabeled: put plate on rack
Policy: push the top drawer
Relabeled: pick up blue shirt
Post-training

IndoorUAV:

Benchmarking Vision-Language UAV Navigation in Continuous Indoor Environments

Indoor Ground-based Agents

- Confined to 2D navigation
- Have limitations to reason about or interact with the full 3D structure of the space

Outdoor UAV

- Lack dense obstacles
- Lack high-precision maneuvering requirements

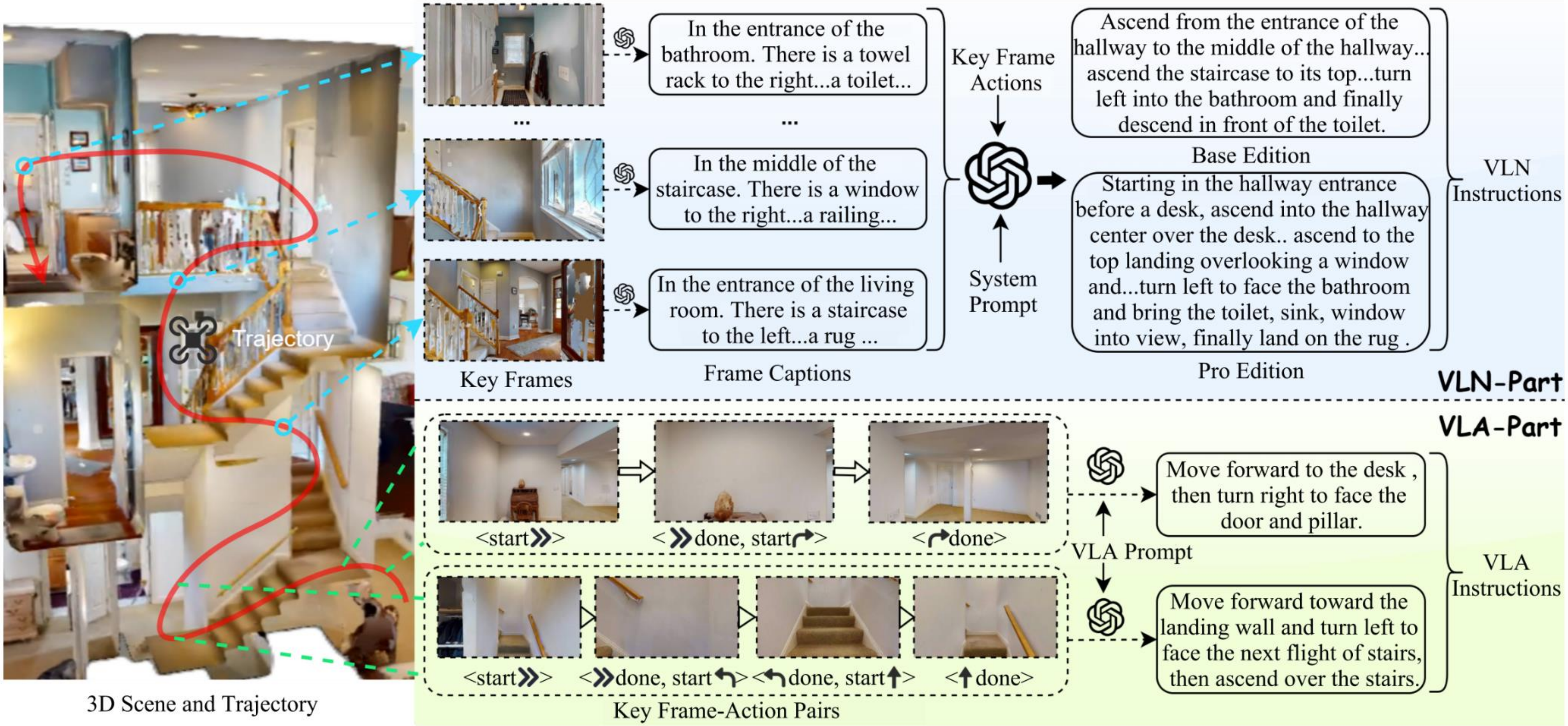
Real-world indoor application

- demand vertical reasoning, free-form 3D maneuvering, and fine-grained spatial understanding

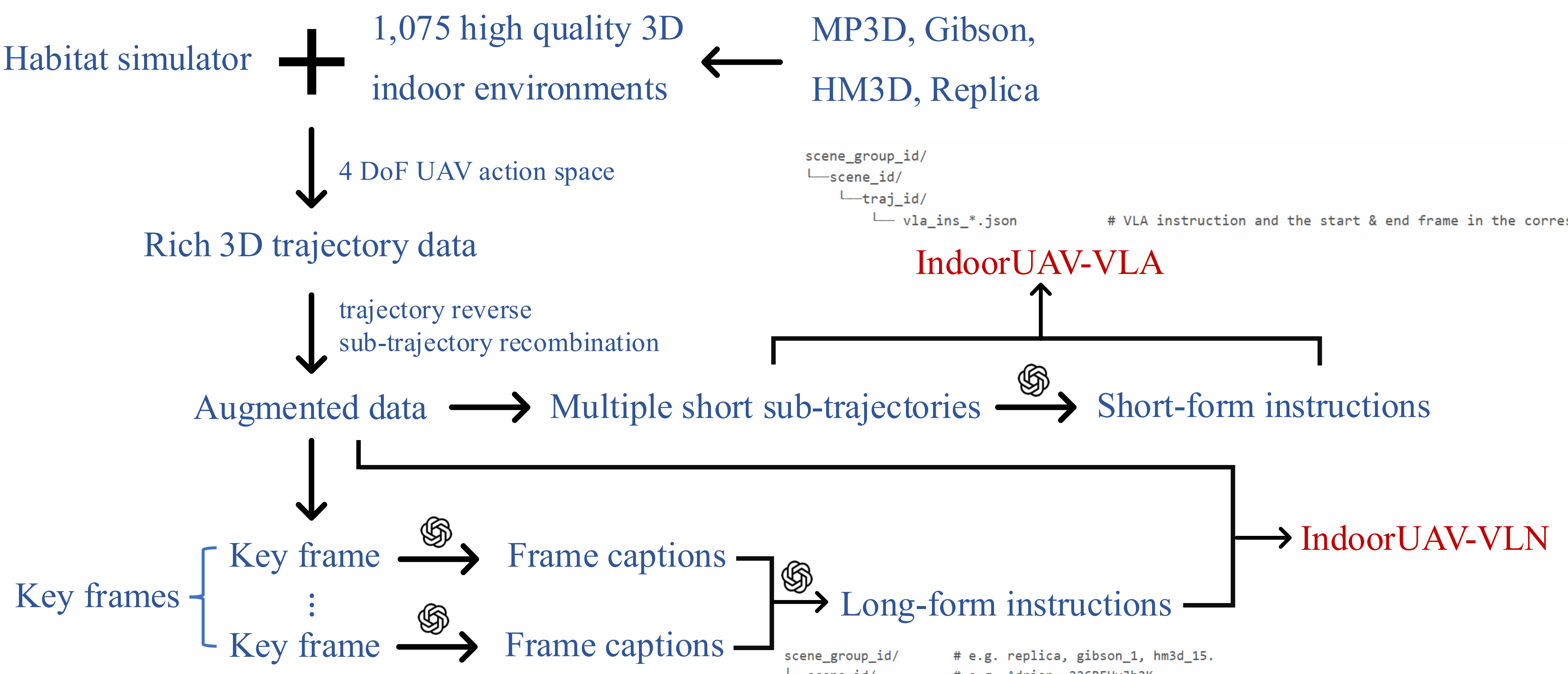
Motivation of IndoorUAV：

- high-level instruction grounding
- low-level aerial control

Data collection



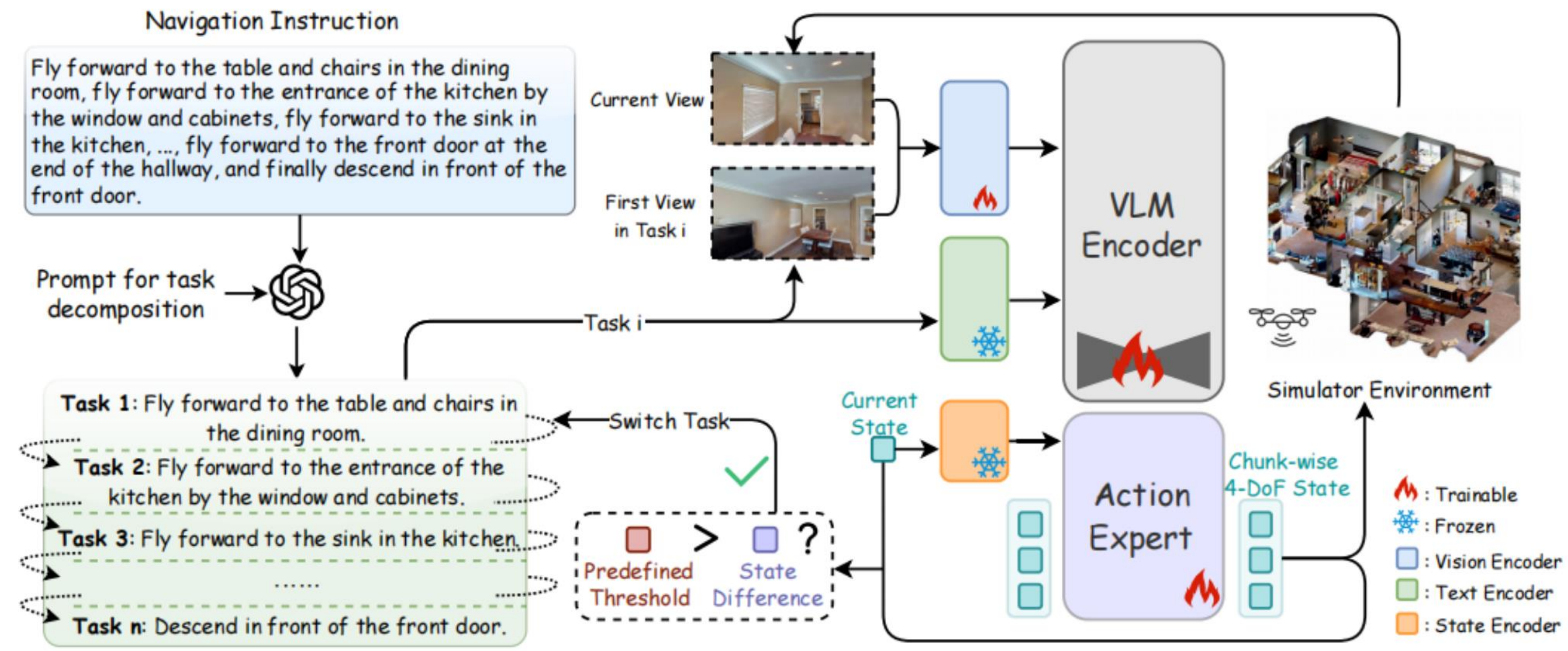
Data collection




```
scene_group_id/
└─scene_id/
    └─traj_id/
        └── vla_ins_*.json          # VLA instruction and the start & end frame in the corresponding VLN trajctory.
```

```
scene_group_id/       # e.g. replica, gibson_1, hm3d_15.
└─scene_id/           # e.g. Adrian, 226REUyJh2K.
   └─traj_id/         # Trajectory collected from the scene; the traj_-* is the inverse edtion of the corresponding
traj_*.
         ├── screenshots/              # The 720P, 10FPS camera view in the trajectory.
         └── saved_transformations/    # The raw posture matrix obtained from habitat-sim.
         ├── posture.txt               # The formatted posture information, equivalent to saved_transformations.
         ├── real_action.json          # The actual movement type and displacement magnitude per frame.
         ├── annotation.json           # Frame numbers annotated with standard action labels.
         ├── key_frames.json           # Key frame numbers and their corresponding descriptive captions.
         ├── instruction.json          # Simplified edition of the natural language instruction.
         └── instruction_pro.json      # Detailed edition of the natural language instruction.
```

Model architecture



Long-horizon tasks ← IndoorUAV-Agent { Task decomposition; Fine-tuned π_0 → Short-horizon tasks }

Evaluation

IndoorUAV-VLA	Full		Easy		Medium		Hard	
	SR/%↑	NDTW/%↑	SR/%↑	NDTW/%↑	SR/%↑	NDTW/%↑	SR/%↑	NDTW/%↑
GPT-4o	11.69	9.2	30.30	12.30	4.00	6.57	1.96	4.84
Seq2Seq*	1.33	2.74	1.6	2.63	1.2	2.74	1.03	3.03
CMA*	0.99	1.88	1.28	1.75	0.75	1.94	1.03	2.01
OpenVLA*	7.81	2.42	22.52	2.89	1.19	1.12	0.0	0.12
π_0-FAST*	8.62	4.71	18.09	8.83	5.26	2.93	1.14	2.68
NaVid*	15.82	5.28	25.31	13.1	18.21	3.21	2.31	1.72
π_0*	27.16	9.44	46.58	14.52	21.64	7.64	7.55	4.27

Table 2: Comparison results on the IndoorUAV-VLA test split. * indicates models fine-tuned on the dataset.

- Easy: 1 action type
- Medium: 2 action types
- Hard: 3 action types

IndoorUAV-VLN	TEST SEEN				TEST UNSEEN			
	NE/m↓	SR/%↑	OSR/%↑	NDTW/%↑	NE/m↓	SR/%↑	OSR/%↑	NDTW/%↑
GPT-4o	7.96	3.69	4.67	6.96	8.52	0.56	2.78	6.82
Seq2Seq*	10.18	0.89	2.23	1.41	11.6	0.41	1.85	1.23
CMA*	11.6	1.56	8.48	1.39	12.15	1.64	10.47	0.81
π_0*	8.35	2.92	11.5	11.87	8.81	2.83	10.02	11.69
NaVid*	21.83	0.75	14.70	1.36	19.40	0.84	16.21	2.32
OpenFly-Agent*	8.17	4.12	10.96	10.63	8.83	2.58	9.45	10.14
Ours	6.62	7.29	12.83	17.19	7.27	5.06	13.49	15.65

Table 3: Comparison results on the IndoorUAV-VLN test split. * indicates models fine-tuned on the dataset.

- Success Rate (SR)
- Normalized Dynamic Time Warping (NDTW)
- Navigation Error (NE)
- Oracle Success Rate (OSR)

Thanks for listening!

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2026.02.05